



IN241 Statistika

Dosen:

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Komponen Penilaian

Komponen-komponen Nilai Akhir:

- Ujian Tengah Semester (UTS)
- Ujian Akhir Semester (UAS)
- Kegiatan Akademik Terstruktur (KAT)

Nilai Akhir (0-100) merupakan **kewenangan dosen** dan ditentukan berdasarkan:

$$N_A = 20\%UTS + 20\%UAS + 60\%KAT$$

Konversi Nilai Akhir (0-100) ke huruf (A,B+,B,C+,C,D,E) merupakan **kewenangan fakultas** dan **dilakukan oleh sistem**.

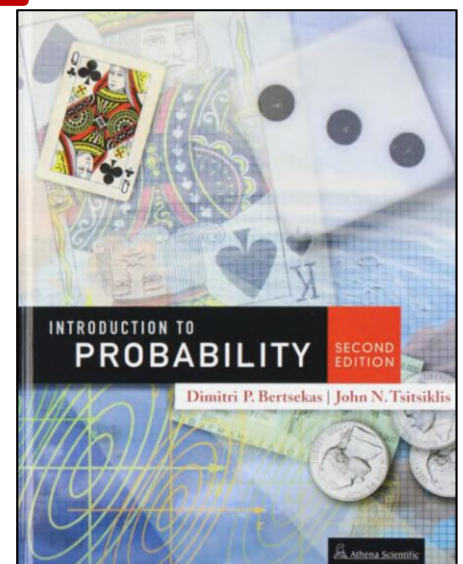
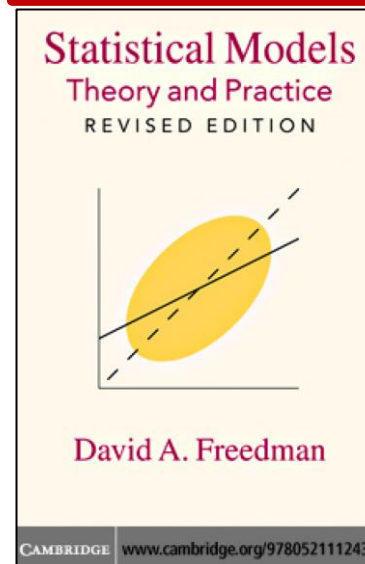
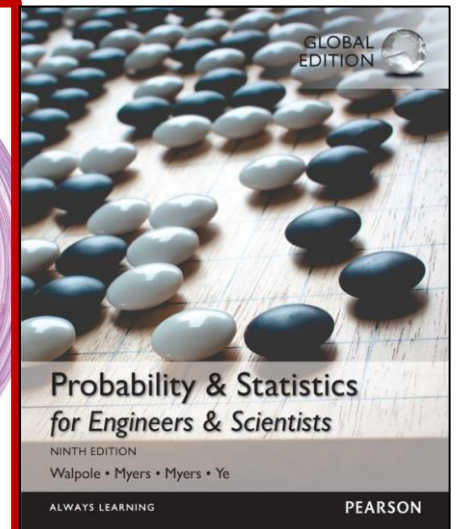
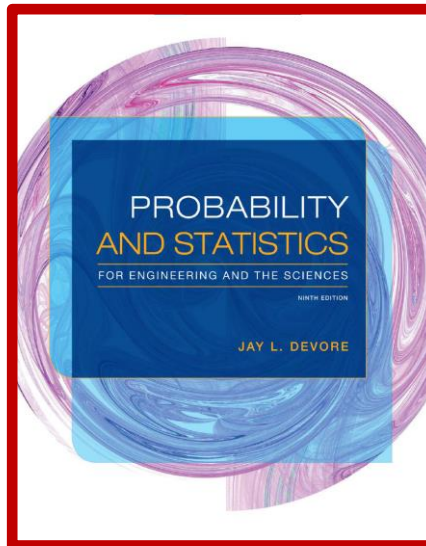
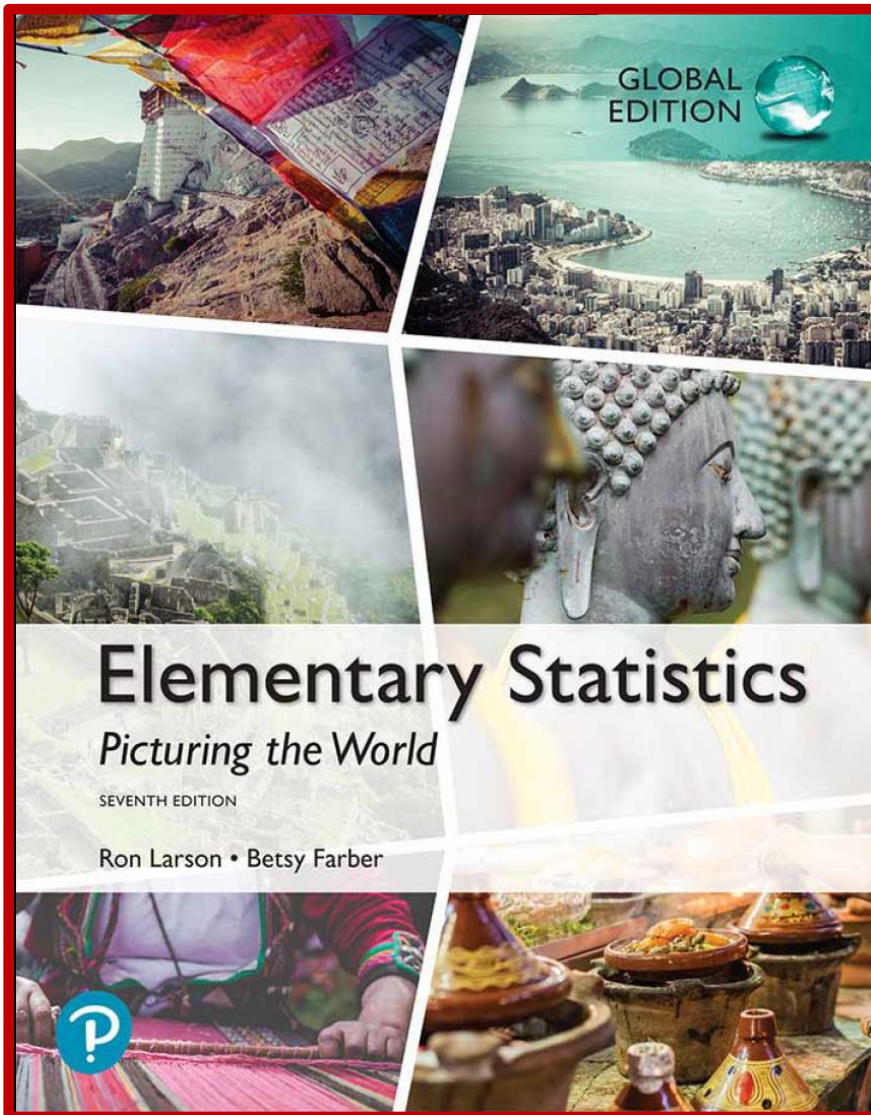
Kegiatan Akademik Terstruktur (KAT)

- PR, Tugas, Quiz
- Presentasi mahasiswa
- Keaktifan mahasiswa
- Partisipasi mahasiswa

Referensi

1. Larson, R. and Farber, B. (2019). *Elementary Statistics: Picturing the World, 7th Edition, Global Edition*, Pearson Education Limited. (Referensi Utama)
2. Devore, J. (2016). *Probability and Statistics for Engineering and the Sciences, 9th Edition*, Brooks/Cole Cengage Learning. (Referensi Utama)
3. Walpole, W., Myers, R.H., Myers, S.L., and Ye, K. (2016). *Probability & Statistics for Engineers & Scientists, 9th Edition, Global Edition*, Pearson Education Limited.
4. Freedman, D.A. (2009). *Statistical Models: Theory and Practice Revised Edition*, Cambridge University Press.
5. Bertsekas, D.P., Tsitsiklis, J.N. (2008) *Introduction to Probability, 2nd Edition*, Athena Scientific.

Referensi



Materi Perkuliahan

- Introduction to Statistics
- Descriptive Statistics
- Probability
- Random Variables and Expectation
- Special Random Variables and Probability Distributions
- Estimation: Confidence Interval
- Hypothesis Testing
- Simple Linear Regression
- Non-Linear Regression and Linearization
- Analysis of Variance (ANOVA)
- Additional Topics (Multi-linear Regression, *etc.*)



IN241 Statistika

Lecture 01

Introduction to Statistics

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An Overview of Statistics

*Based on textbooks of Larson (2019), Ch. 1.

An Overview of Statistics

Definition:

Data consist of information coming from observations, counts, measurements, or responses.

Definition:

Statistics is the science of collecting, organizing, analyzing, and interpreting data in order to make decisions.

An Overview of Statistics

Data Sets

There are two types of data sets you will use when studying statistics. These data sets are called **populations** and **samples**.

Definition:

A **population** is the collection of all outcomes, responses, measurements, or counts that are of interest. A **sample** is a subset, or part, of a population.

An Overview of Statistics

A sample is used to gain information about a population. A sample should be representative of a population so that sample data can be used to draw conclusions about that population. Sample data must be collected using an appropriate method, such as *random sampling*. When sample data are collected using an *inappropriate* method, the data cannot be used to draw conclusions about the population.

Whether a data set is a population or a sample usually depends on the context of the real-life situation.

An Overview of Statistics

Definition:

A **parameter** is a numerical description of a *population* characteristic.

A **statistic** is a numerical description of a *sample* characteristic.

It is important to note that a sample statistic can differ from sample to sample, whereas a population parameter is constant for a population.

Branches of Statistics

The study of statistics has two major branches: **descriptive statistics** and **inferential statistics**.

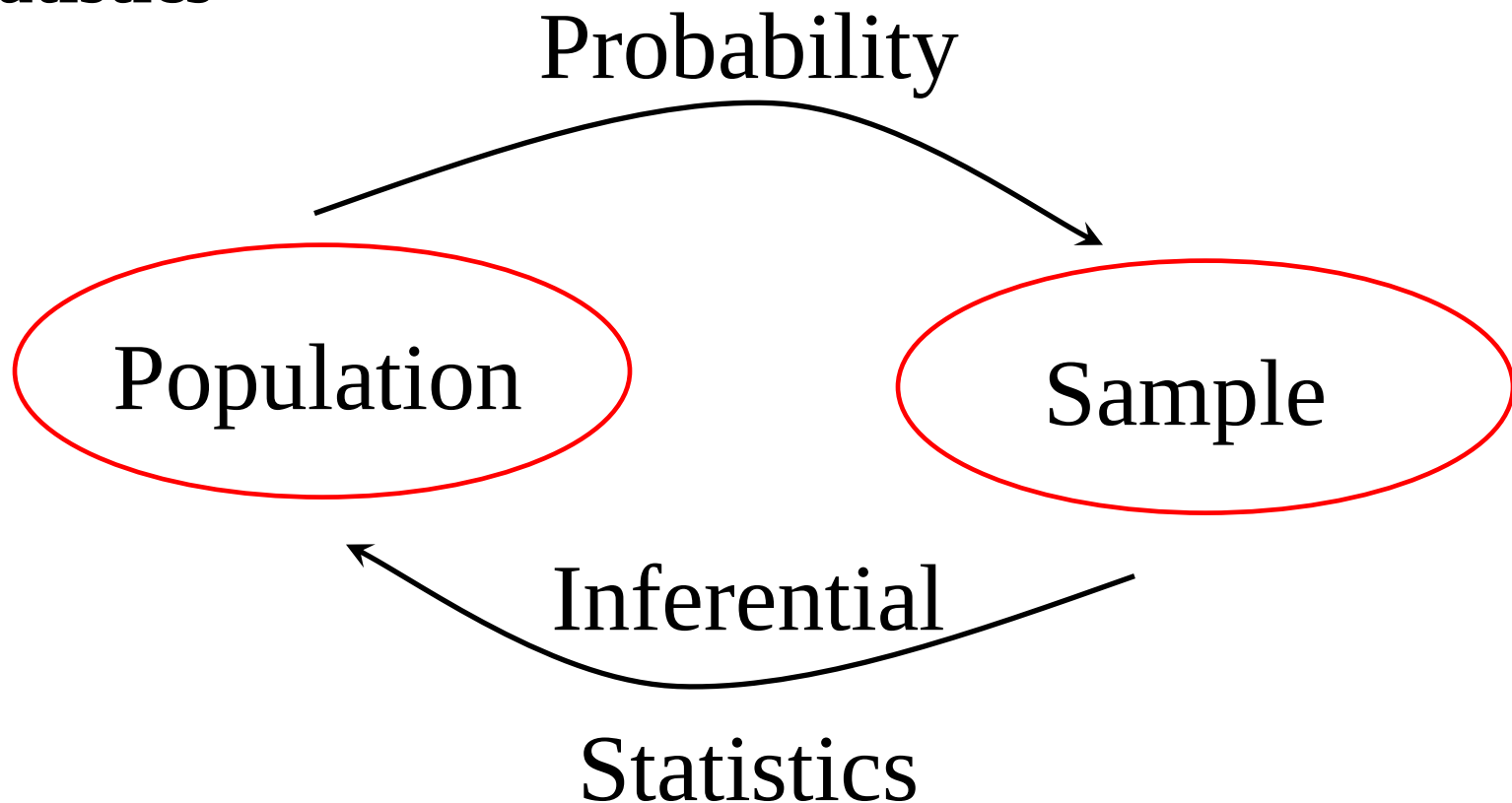
Definition:

Descriptive statistics is the branch of statistics that involves the organization, summarization, and display of data.

Inferential statistics is the branch of statistics that involves using a sample to draw conclusions about a population. A basic tool in the study of inferential statistics is *probability*.

Branches of Statistics

Relationship Between **Probability** and **Inferential Statistics**



Data Classification

Types of Data

When conducting a study, it is important to know the kind of data involved. The type of data you are working with will determine which statistical procedures can be used. Data sets can consist of two types of data: **qualitative data** and **quantitative data**.

Definition:

Qualitative data consist of attributes, labels, or nonnumerical entries.

Quantitative data consist of numbers that are measurements or counts.

Data Classification

Levels of Measurement

Another characteristic of data is its level of measurement. The level of measurement determines which statistical calculations are meaningful. The four levels of measurement, in order from lowest to highest, are **nominal**, **ordinal**, **interval**, and **ratio**.

Definition:

Data at the **nominal level of measurement** are qualitative only. Data at this level are categorized using names, labels, or qualities. No mathematical computations can be made at this level.

Data at the **ordinal level of measurement** are qualitative or quantitative. Data at this level can be arranged in order, or ranked, but differences between data entries are not meaningful.

Data Classification

Levels of Measurement

The two highest levels of measurement consist of quantitative data only.

Definition:

Data at the **interval level of measurement** can be ordered, and meaningful differences between data entries can be calculated. At the interval level, a zero entry simply represents a position on a scale; the entry is not an *inherent zero*.

Data at the **ratio level of measurement** are similar to data at the interval level, with the added property that a zero entry is an *inherent zero*. A ratio of two data entries can be formed so that one data entry can be meaningfully expressed as a multiple of another.

Data Classification

Levels of Measurement

An **inherent zero** is a zero that implies “none.” For instance, the amount of money you have in a savings account could be zero dollars. In this case, the zero represents no money; it is an inherent zero. On the other hand, a temperature of 0°C does not represent a condition in which no heat is present. The 0°C temperature is simply a position on the Celsius scale; it is not an inherent zero.

To distinguish between data at the interval level and at the ratio level, determine whether the expression “twice as much” has any meaning in the context of the data. For instance, \$2 is twice as much as \$1, so these data are at the ratio level. On the other hand, 2°C is not twice as warm as 1°C , so these data are at the interval level.

Data Classification

Levels of Measurement

The tables below summarize which operations are meaningful at each of the four levels of measurement. When identifying a data set's level of measurement, use the highest level that applies.

Level of measurement	Put data in categories	Arrange data in order	Subtract data entries	Determine whether one data entry is a multiple of another
Nominal	Yes	No	No	No
Ordinal	Yes	Yes	No	No
Interval	Yes	Yes	Yes	No
Ratio	Yes	Yes	Yes	Yes

Summary of Four Levels of Measurement

	Example of a data set	Meaningful calculations
Nominal level (Qualitative data)	<i>Types of Shows Televised by a Network</i> Comedy Documentaries Drama Cooking Reality Shows Soap Operas Sports Talk Shows	<i>Put in a category.</i> For instance, a show televised by the network could be put into one of the eight categories shown.
Ordinal level (Qualitative or quantitative data)	<i>Motion Picture Association of America Ratings Description</i> G General Audiences PG Parental Guidance Suggested PG-13 Parents Strongly Cautioned R Restricted NC-17 No One 17 and Under Admitted	<i>Put in a category and put in order.</i> For instance, a PG rating has a stronger restriction than a G rating.
Interval level (Quantitative data)	<i>Average Monthly Temperatures (in degrees Fahrenheit) for Denver, CO</i> Jan 30.7 Jul 74.2 Feb 32.5 Aug 72.5 Mar 40.4 Sep 63.4 Apr 47.4 Oct 50.9 May 57.1 Nov 38.3 Jun 67.4 Dec 30.0 <i>(Source: National Climatic Data Center)</i>	<i>Put in a category, put in order, and find differences between data entries.</i> For instance, $72.5 - 63.4 = 9.1^{\circ}\text{F}$. So, August is 9.1°F warmer than September.
Ratio level (Quantitative data)	<i>Average Monthly Precipitation (in inches) for Orlando, FL</i> Jan 2.35 Jul 7.27 Feb 2.38 Aug 7.13 Mar 3.77 Sep 6.06 Apr 2.68 Oct 3.31 May 3.45 Nov 2.17 Jun 7.58 Dec 2.58 <i>(Source: National Climatic Data Center)</i>	<i>Put in a category, put in order, find differences between data entries, and find ratios of data entries.</i> For instance, $\frac{7.58}{3.77} \approx 2.$ So, there is about twice as much precipitation in June as in March.

Data Collection & Experimental Design

Guidelines of Designing a Statistical Study

1. Identify the variable(s) of interest (the focus) and the population of the study.
2. Develop a detailed plan for collecting data. If you use a sample, make sure the sample is representative of the population.
3. Collect the data.
4. Describe the data, using descriptive statistics techniques.
5. Interpret the data and make decisions about the population using inferential statistics.
6. Identify any possible errors.

A **statistical study** can usually be categorized as an **observational study** or an **experiment**. In an **observational study**, a researcher does not influence the responses. In an **experiment**, a researcher deliberately applies a treatment before observing the responses.

Data Collection

A **simulation** is the use of a mathematical or physical model to reproduce the conditions of a situation or process. Collecting data often involves the use of computers. Simulations allow you to study situations that are impractical or even dangerous to create in real life, and often they save time and money. For instance, automobile manufacturers use simulations with dummies to study the effects of crashes on humans.

A **survey** is an investigation of one or more characteristics of a population. Most often, surveys are carried out on people by asking them questions. The most common types of surveys are done by interview, Internet, phone, or mail. In designing a survey, it is important to word the questions so that they do not lead to biased results, which are not representative of a population. For instance, a survey is conducted on a sample of female physicians to determine whether the primary reason for their career choice is financial stability. In designing the survey, it would be acceptable to make a list of reasons and ask each individual in the sample to select her first choice.

Experimental Design

Definition:

A **confounding variable** occurs when an experimenter cannot tell the difference between the effects of different factors on the variable.

Definition:

Blinding is a technique where the subjects do not know whether they are receiving a treatment or a placebo. In a **double-blind experiment**, neither the experimenter nor the subjects know whether the subjects are receiving a treatment or a **placebo**. The experimenter is informed after all the data have been collected. This type of experimental design is preferred by researchers.

Definition:

Randomization is a process of randomly assigning subjects to different treatment groups.

Definition:

Replication is the repetition of an experiment under the same or similar conditions.

Sampling Technique

A **census** is a count or measure of an entire population. Taking a census provides complete information, but it is often costly and difficult to perform. A **sampling** is a count or measure of part of a population and is more commonly used in statistical studies. To collect unbiased data, a researcher must ensure that the sample is representative of the population. Appropriate sampling techniques must be used to ensure that inferences about the population are valid. Remember that when a study is done with faulty data, the results are questionable. Even with the best methods of sampling, a *sampling error* may occur. A **sampling error** is the difference between the results of a sample and those of the population.

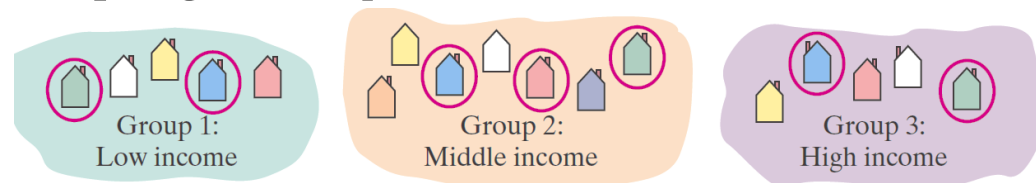
A **random sample** is one in which every member of the population has an equal chance of being selected (randomly). A simple random sample is a sample in which every possible sample of the same size has the same chance of being selected.

Sampling Technique

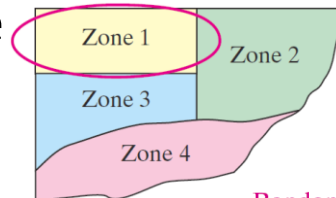
When you choose members of a sample, you should decide whether it is acceptable to have the same population member selected more than once. If it is acceptable, then the sampling process is said to be **with replacement**. If it is not acceptable, then the sampling process is said to be **without replacement**.

Other commonly used sampling techniques:

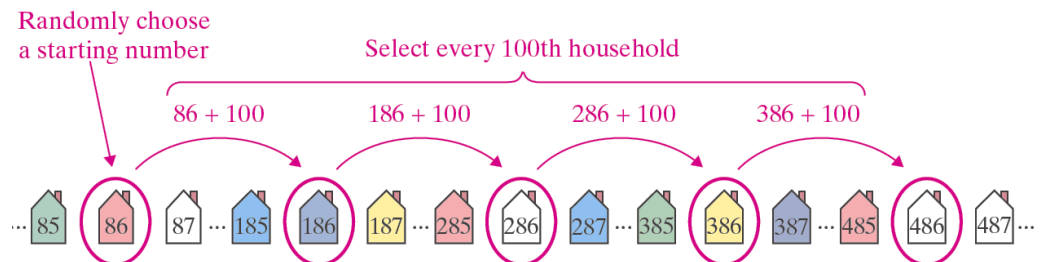
- **Stratified Sample**



- **Cluster Sample**



- **Systematic Sample**





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